

Foundation (Jalapeno)

- Q1.** What is the factored form of $x^3 + 8$?
(A) $(x + 2)(x^2 - 2x + 4)$
(B) $(x - 2)(x^2 + 2x + 4)$
(C) $(x + 2)(x^2 - 4x + 4)$
(D) $(x - 2)(x^2 + 4x + 4)$
(E) $(x + 2)(x^2 + 2x + 4)$
- Q2.** Factor the expression $27x^3 - 64$
(A) $(3x - 4)(9x^2 + 12x + 16)$
(B) $(3x + 4)(9x^2 - 12x + 16)$
(C) $(3x - 2)(9x^2 + 6x + 4)$
(D) $(3x + 2)(9x^2 - 6x + 4)$
(E) $(3x - 2)(9x^2 + 6x + 4)$
- Q3.** A company's stock value is given by $x^3 - 125$. What is the factored form?
(A) $(x - 5)(x^2 + 5x + 25)$
(B) $(x + 5)(x^2 - 5x + 25)$
(C) $(x - 5)(x^2 - 5x + 25)$
(D) $(x + 5)(x^2 + 5x + 25)$
(E) $(x + 5)(x^2 - 5x + 25)$
- Q4.** A taxpayer's savings can be modeled by $1000x^3 + 8$. Factor this expression
(A) $2(5x + 2)(10x^2 - 10x + 4)(5)$
(B) $2(5x - 2)(10x^2 + 10x + 4)(5)$
(C) $4(5x - 2)(25x^2 + 10x + 2)$
(D) $4(5x + 2)(25x^2 - 10x + 2)$
(E) $2(5x - 2)(10x^2 - 10x + 4)$
- Q5.** What is the factored form of $x^3 - 64y^3$?
(A) $(x - 4y)(x^2 + 4xy + 16y^2)$
(B) $(x + 4y)(x^2 - 4xy + 16y^2)$
(C) $(x - 8y)(x^2 + 8xy + 64y^2)$
(D) $(x + 8y)(x^2 - 8xy + 64y^2)$
(E) $(x - 2y)(x^2 + 2xy + 4y^2)$
- Q6.** The cost of producing a good is given by $x^3 + 27$. Factor this expression
(A) $(x + 3)(x^2 - 3x + 9)$
(B) $(x - 3)(x^2 + 3x + 9)$
(C) $3(x + 3)(x^2 - 3x + 3)$
(D) $3(x - 3)(x^2 + 3x + 3)$
(E) $3(x + 1)(x^2 - 3x + 3)$
- Q7.** A financial analyst's data is given by $125x^3 - 8y^3$. What is the factored form?
(A) $(5x - 2y)(25x^2 + 10xy + 4y^2)$
(B) $(5x + 2y)(25x^2 - 10xy + 4y^2)$
(C) $(5x - 2y)(25x^2 + 10xy + 4y^2)$
(D) $(5x + 2y)(25x^2 + 10xy + 4y^2)$
(E) $(5x + 2y)(25x^2 - 10xy + 4y^2)$
- Q8.** The price of a product can be modeled by $x^3 - 1$. Factor this expression
(A) $(x - 1)(x^2 + x + 1)$
(B) $(x + 1)(x^2 - x + 1)$
(C) $(x - 1)(x^2 - x + 1)$
(D) $(x + 1)(x^2 + x + 1)$
(E) $(x + 2)(x^2 + x + 1)$

Open Ended Questions (FRQ)

- Q9.** Factor the expression $64x^3 + 27y^3$, showing all steps and simplifying fully.
- Q10.** A company's profit can be modeled by $x^3 - 1000$. Factor this expression and explain the meaning of each factor in the context of the problem.

Chef's Tips

- Tip 1: Focus on identifying cube patterns
- Tip 2: Use the sum and difference of cubes formulas
- Tip 3: Apply the formulas to real-world problems in business and finance